

SKILL – 14:

PART 1 — Append Mode:

Code:

```
GNU nano 7.2 skill14_append.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {
    int fd;
    const char *file = "append_data.txt";

    // Open file in append mode
    fd = open(file, O_WRONLY | O_CREAT | O_APPEND, 0644);

    if (fd == -1) {
        perror("Error opening file");
        return 1;
    }

    const char *message = "New data appended to the file.\n";

    if (write(fd, message, strlen(message)) == -1) {
        perror("Error writing to file");
        close(fd);
        return 1;
    }

    close(fd);
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ nano skill14_append.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ gcc skill14_append.c -o skill14_append
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ ./skill14_append
Data appended successfully.

Current file contents:
New data appended to the file.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ |
```

Run it again to prove APPEND:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ ./skill14_append
Data appended successfully.

Current file contents:
New data appended to the file.
New data appended to the file.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ |
```

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ cat append_data.txt
New data appended to the file.
New data appended to the file.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ |
```

PART 2 — stderr REDIRECTION:

Code:

```
GNU nano 7.2 skill14_stderr.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    printf("Testing stderr redirection...\n");

    fd = open("file_that_does_not_exist.txt", O_RDONLY);

    if (fd == -1) {
        perror("ERROR");
        return 1;
    }

    close(fd);

    return 0;
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ gcc skill14_stderr.c -o skill14_stderr
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ ./skill14_stderr
Testing stderr redirection...
ERROR: No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ |

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ ./skill14_stderr 2> error_output.txt
Testing stderr redirection...
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ cat error_output.txt
ERROR: No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ |
```

Test stderr APPEND 2>>:

```
Testing stderr redirection...
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ cat error_output.txt
ERROR: No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ ./skill14_stderr 2>> error_output.txt
Testing stderr redirection...
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ cat error_output.txt
ERROR: No such file or directory
ERROR: No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ |
```

Verify the files:

```
ERROR: No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ ls -l append_data.txt error_output.txt
-rw-r--r-- 1 bhardwaj bhardwaj 62 Oct  7 16:34 append_data.txt
-rw-r--r-- 1 bhardwaj bhardwaj 66 Oct  7 16:37 error_output.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ |
```

Test both stdout and stderr:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ ./skill14_stderr > output.txt 2> errors.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ cat output.txt
Testing stderr redirection...
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ ^C
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ cat errors.txt
ERROR: No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-14$ |
```